Nathan McNew

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Employment

Towson University

Associate Professor — 2021-Present
Assistant Professor — 2015-2021
Fisher Endowed Chair in Mathematics — December 2016-June 2019

Education

Dartmouth College — Fall 2010 - Spring 2015

Ph.D. Mathematics, Advisor: Carl Pomerance — June 2015
Thesis: Multiplicative problems in combinatorial number theory

A.M. Mathematics — June 2012

University of Denver — Fall 2006 - Spring 2010

B.S. Mathematics and Physics, Summa Cum Laude, Phi Beta Kappa — June 2010

Publications

1. **Matchable Numbers**

   With C. Pomerance. *Mathematika.* **72** (2026). https://doi.org/10.1112/mtk.70116

2. **Iterated harmonic numbers**

   With J Marshall Ash, Michael Ash, Rafael Ash, and Angel Plaza. *College Math. J.* **57** (2026), no. 3, 193–205.

3. **On the densities of covering numbers and abundant numbers**

   With J. Setty. To appear in *Mathematics of Computation.*

4. **Explicit bounds for large gaps between cubefree integers**

   With A. Kumchev, W. McCormick, A. Park, R. Scherr and S. Ziehr. *Combinatorial and Additive Number Theory VI* (Proceedings of CANT: May 2022 and 2023), Springer Proceedings in Mathematics & Statistics, **464** (2025) 259–282. https://doi.org/10.1007/978-3-031-65064-2_14

5. **Short interval results for powerfree polynomials over finite fields**

   With A. Kumchev, and A. Park. *International Journal of Number Theory.* **20** (2024) 867–892. https://doi.org/10.1142/S1793042124500453

6. **Links and the Diaconis Graham inequality**

   With C. Cornwell. *Combinatorica.* **44** (2024) 1149–1167. https://doi.org/10.1007/s00493-024-00107-1

7. **The distribution of intermediate prime factors**

   With P. Pollack, A. Singha Roy. *Illinois J. Math.* **68 (3)** (2024) 537–576. http://dx.doi.org/10.1215/00192082-11417186

8. **Intermediate prime factors in specified subsets**

   With P. Pollack, A. Singha Roy. *Monatshefte für Mathematik*. **202** (2023), 837–855. https://doi.org/10.1007/s00605-023-01855-w

9. **Explicit bounds for large gaps between squarefree integers**

With A. Kumchev, W. McCormick, A. Park, R. Scherr and S. Ziehr. *Journal of Number Theory.* **254** (2024), 336–357. `https://doi.org/10.1016/j.jnt.2023.07.003`

10. **Permutations and the divisor graph of** $[1, n]$

    Mathematika. **69 (1)** (2023), 51–67. `https://doi.org/10.1112/mtk.12177`

11. **Note on sets without geometric progressions**

    With X. Cao, J. Fang. *Integers.* **22** (2022), #A45. `http://math.colgate.edu/~integers/w45/w45.pdf`

12. **Unknotted Cycles**

    With C. Cornwell. *The Electronic Journal of Combinatorics.* **29 (3)** (2022), #P3.50 `https://doi.org/10.37236/11016`

13. **On the Erdős primitive set conjecture in function fields**

    With A. Gomez-Colunga, C. Kavaler and M. Zhu. *Journal of Number Theory.* **219** (2021), 412–444. `https://doi.org/10.1016/j.jnt.2020.09.001`

14. **On the size of primitive sets in function fields**

    With A. Gomez-Colunga, C. Kavaler and M. Zhu. *Finite Fields and their Applications.* **64** (2020), 101658. `https://doi.org/10.1016/j.ffa.2020.101658`

15. **Counting pattern-avoiding integer partitions**

    With J. Bloom. *The Ramanujan Journal.* **55** (2021) 555–591.
    `https://doi.org/10.1007/s11139-020-00287-6`

16. **Counting primitive sets and other statistics of the divisor graph of $\{\mathbf{1}, \mathbf{2}, \ldots \mathbf{n}\}$**

    The European Journal of Combinatorics. **92** (2021) 103237.
    `https://doi.org/10.1016/j.ejc.2020.103237`

17. **Primitive and geometric-progression-free sets without large gaps**

    Acta Arithmetica. **192** (2020), 95–104. `https://doi.org/10.4064/aa180921-4-2`

18. **Avoiding 3-Term Geometric Progressions in Hurwitz Quaternions**

    With M. Asada, B. Fang, E. Fourakis, S. Manski, S. J. Miller, G. Moreland, A. Yamin and S. Zhang, *Journal of Integer Sequences.* **27** (2024) 24.7.7.

19. **When sets can and cannot have sum-dominant subsets**

    With H. Chu, S. J. Miller, V. Xu and S. Zhang, *Journal of Integer Sequences.* **21** (2018) 18.8.2.

20. **The convex hull of the prime number graph**

    In *Irregularities in the Distribution of Prime Numbers* Pintz J., Rassias M. (eds) Springer, Cham. 2018, pp. 125–141.

21. **Random multiplicative walks on the residues modulo n**

    Mathematika. **63** (2017), 602–621.

22. **Ramsey theory problems over the integers: avoiding generalized progressions**

    With A. Best, K. Huan, S. J. Miller, J. Powell, K. Tor and M. Weinstein. In *Combinatorial and Additive Number Theory II. CANT 2015, 2016.* Nathanson, M. (eds) Springer Proceedings in Mathematics & Statistics. **220**. Springer, New York, NY, 2017, pp. 39–52.

23. **Numbers divisible by a large shifted prime and large torsion subgroups of CM elliptic curves**

    With P. Pollack and C. Pomerance. *International Math Research Notices.* **18** (2017), 5525–5553.

24. **The most frequent values of the largest prime divisor function**

    Experimental Mathematics. **26** (2017), 210–224.

25. **Subsets of $\mathbb{F}_q[x]$ free of 3-term geometric progressions**

    With M. Asada, E. Fourakis, S. Manski, S. J. Miller and G. Moreland. *Finite Fields and Their Applications.* **44** (2017), 135–147.

26. **Geometric-progression-free sets over quadratic number fields**

    With A. Best, K. Huan, S. J. Miller, J. Powell, K. Tor and M. Weinstein. *Proceedings of the Royal Society of Edinburgh Section A.* **147** (2017), 245–262.

27. **Infinitude of k-Lehmer numbers which are not Carmichael**
    With T. Wright. *International Journal of Number Theory*. **12** (2016), 1863–1869.
28. **On sets of integers which contain no three terms in geometric progression**
    Mathematics of Computation. **84** (2015), 2893–2910.
29. **Efficient realization of nonzero spectra by polynomial matrices**
    With N. Ormes. *Involve, A Journal of Mathematics.* **8** (2015), 1–24.
30. **Radically weakening the Lehmer and Carmichael conditions**
    International Journal of Number Theory. **9** (2013), 1215–1224.

Awards and Honors and Grants

FCSM Mentoring Award — Fall 2024

REU Grant for Towson University Math REU CO-PI, $100,000 Award, NSA — 2020

REU Grant for Towson University Math REU Key Personnel, $300,000 Award, NSF — 2022-2024

S-STEM Grant to support math majors at TU Key Personnel, $1,000,000 Award, NSF — 2021-2024

Grant to adapt OpenStax for Calculus I PI, Maryland Open Source Textbook Initiative — Fall 2018

Grant for MASON II-IV Co-PI, NSF — 2018-2020

Jess and Mildred Fisher Endowed Chair in the Mathematical and Computing Sciences
Towson University — 2016-2019

Grant for MASON I Number Theory Foundation — October 2016

Dartmouth Graduate Poster Session Winner — Spring 2015

Outstanding Graduate Student Teaching Award — April 2014
Dartmouth Center for the Advancement of Learning

Teaching

Associate Professor, Towson University

MATH 273: Calculus I	Fall 2021, Spring 2024, Spring 2025, Fall 2025
MATH 314: Cryptography	Fall 2021, Spring 2022 (×2), Fall 2023 (×2), Spring 2024, Fall ’24
MATH 369: Introduction to Abstract Algebra	Fall 2025
MATH 378: Experimental Mathematics	Fall 2024
MATH 475: Complex Analysis	Spring 2025
MATH 491: Readings in Mathematics	Spring 2025
IDNM 400: Exploration of Careers in Mathematics	Fall 2021
IDNM 200: Exploration of Careers in Mathematics	Fall 2023
ORIE 101: Student support for Calculus I	Fall 2021
COSC 495: Independent Study (Quantum Computing)	Spring 2022

Assistant Professor, Towson University

Math 273: Calculus I	Fall 2015, Fall 2016, Fall 2018, Fall 2020
Math 275: Calculus III	Spring 2016 (×2)
Math 314: Cryptography	Fall 2016, Spring 2017, Fall 2017, Spring 2018 (×2), Fall 2018 Spring 2019, Fall 2019 (×2), Spring 2020 (×2), Spring 2021 (×2)
Math 315: Applied Combinatorics	Fall 2017
Math 374: Differential Equations	Spring 2020
Math 378: Experimental Mathematics	Fall 2020

Math 451: Graph Theory Spring 2018
Math 467: Algebraic Structures Fall 2015
Math 490: Senior Seminar Spring 2017
Math 465/565: Theory of Numbers Spring 2019, Fall 2019, Spring 2021

Instructor, Dartmouth College
Math 10: Introduction to Statistics Spring 2013
Math 20: Discrete Probability Fall 2013

Co-Instructor, Dartmouth College
Math 25: Elementary Number Theory Fall 2014

Teaching Assistant, Dartmouth College
Math 8: Calculus of Functions of One and Several Variables Fall 2010
Math 13: Calculus of Vector Valued Functions Fall 2011
Math 23: Differential Equations Winter 2011, Spring 2012

Professional Activities

Visiting Scholar, University of Georgia Fall 2022

Park City Math Institute Summer 2022
Participant in the Undergraduate Faculty Program, a 3-week long lecture and problem series. Topic: Fourier Analysis on Polytopes.

Towson REU Summer 2022
Co-mentoreed, with Angel Kumchev, a group of 4 undergraduate students investigating gaps between squarefree integers and polynomials, with three papers in preparation.

TU REP CURE Course professional development 3rd cohort Spring-Fall 2020
Faculty development program to discuss the design and implementation of Course-based Undergraduate Research Experience courses at TU with an emphasis on inclusive teaching practices.

SUMRY - Summer Undergraduate Math Research at Yale Summer 2019
Led group of undergraduate researchers in combinatorial number theory, resulting in two publications.

MASON II Conference April 2018
Co-organized (with Angel Kumchev) the second in the MASON series of number theory conferences.

Regional Undergraduate Math Conference
Co-organize (with Alexei Kolesnikov, Sergei Borodachov and others) an annual regional undergraduate math conference hosted at Towson for undergraduate students to present research, hear about opportunities for graduate school and network with students from nearby universities.

MASON Conference October 2016
Co-organized (with Angel Kumchev) the first in a new series of regional number theory conferences (the Mid-Atlantic Seminar On Numbers) for the Mid-Atlantic region.

Project NExT August 2016-August 2017
MAA program for new faculty to explore innovative new teaching techniques and transition from graduate school into a teaching position.

MSRI Summer School: Gaps Between Primes July 2015
A two week program with lectures and problem sessions on recent progress on gaps between primes.

Arizona winter school: Arithmetic Statistics March 2014
Workshop with lectures and problem sessions on topics in Arithmetic Statistics. I participated in the problem group for Melanie Matchett Wood's section on asymptotics for number fields and class groups.

Warwick University summer school: number theory for cryptography June 2013

Course for PhD students in number theory and related fields on cryptology. Topics included high-speed cryptography, complex multiplication of elliptic curves, discrete logarithms and integer factorization.

Dartmouth Mathematics Teaching Seminar Summer 2012

An intensive course taken by graduate students who have advanced to PhD candidacy. Involves discussion of educational philosophies, classroom techniques, and course design. Culminates in the design and instruction of two week long math camps for middle and high school students.

Banff International Research Center: Diophantine equations June 2012

A workshop on contemporary techniques in Diophantine equations including the modular approach, the Brauer-Manin obstruction, Chabauty methods, and linear forms in logarithms.

Service

MASON MidAtlantic Seminar On Numbers

- Local Co-Organizer (Towson University) Fall 2016
- Local Co-Organizer (Towson University) Spring 2018
- Co-Organizer (James Madison University) Spring 2019
- Co-Organizer (Gettysburg College) Spring 2020
- Co-Organizer (Online) Spring 2021
- Local Co-Organizer (Towson University) Spring 2025
- Local Co-Organizer (UMD) Spring 2023

Regional Undergraduate Mathematics Conference, Co-organizer 2017, 2018, Spring 2019, Fall 2019, Spring 2021, 2022, 2023, 2024, Fall 2024

Baltimore Combinatorics and Number Theory Seminar Co-organizer Fall 2015-Present

Instructional Information Technology Committee Towson U.

- Fisher College Representative Fall 2025-Present

Undergraduate Research Committee Fisher College, Towson U. 2024-Present

Departmental Honors Coordinator Towson U. 2019-Present

Pure and Applied Mathematics Working Group Towson U.

- Chair Fall 2023-Present

Graduate Committee-Applied and Industrial Math Program Towson U. 2016-2022

Curriculum Committee Towson University Math Department

- Assistant Chair Fall 2019-Spring 2020

Pure Math Committee Towson University Math Department 2015-Spring 2023

- Chair Spring 2018

Department Representative to MAA Towson University Math Department 2016-2023

Colloquium Committee Towson University Math Department 2015-2018, 2021-2022, 2024-Present

- Chair 2017-2018

Math Club Faculty Sponsor Towson University Math Department 2015-2019, Spring 2025

Problem Solving Team Coach Towson University Math Department 2015-Present

Dartmouth Number Theory Seminar Organizer Fall 2011-Spring 2013

Dartmouth Graduate Student Council Math Department Representative Fall 2013-Summer 2014

Referee: 2014-Present

Journal of Integer Sequences, Journal of Number Theory, Information Security Journal, Mathematics Magazine, Mathematics of Computation, Experimental Mathematics, Integers, Mathematics, Symmetry, IEEE Access, Electronic Journal of Combinatorics, Involve, Israel Journal of Mathematics, Mathematika, MAA Monthly, Research in Number Theory, Archiv der Mathematik, College Math Journal

Reviewer: Mathematical Reviews 2015-Present

Educational Outreach

SUMRY (Summer Undergraduate Research at Yale) Project Mentor Summer 2019

Mentored three students through a research project regarding primitive sets of polynomials for a 10 week summer program.

Regional Undergraduate Math Research Conference April 2017, April 2018, April 2019 November 2019, May 2021, April 2022, April 2023, April 2024, November 2024

Co-organized conference for undergraduates in mathematics to present their research and learn about opportunities in graduate school or industry after graduation.

Williams College REU Graduate Mentor Summer 2014, 2015

Worked with undergraduates at the Williams College REU on research in combinatorial number theory.

Putnam Supervisor at Dartmouth College and Towson University 2013-Present

Helped students to prepare for the Putnam competition, and proctored the exam.

Extreme Academics, University of Denver Nov '11, Mar '13, Feb '14, Apr '15, Mar '16, Apr '17

Invited to participate in a panel discussion about applying to and doing research in grad school.

Young Mathematicians Conference, Ohio State University August 2014

Mentor to students and served on a panel discussion about applying to graduate school.

Dartmouth College Science Day April 2014

Showed visiting elementary school students about the mathematics of hexaflexagons.

Johns Hopkins Center for Talented Youth May 2011

Designed and led three workshops for middle and high school students. Topic: Cryptography

Vermont Southeast Regional MATHCOUNTS Volunteer February 2011

Gave a talk to middle school students about math research, helped proctor and grade competition.

MATHCOUNTS Coach: Jefferson Academy Middle School, Broomfield CO 2006-2010

Led weekly problem sessions with the team, discussed problem solving strategies and concepts.

Selected Presentations

Matchable Numbers
- Combinatorial and Additive Number Theory, CUNY — July 2026

Experimental Mathematics as a CURE
- JMM - Session on Course based Undergraduate Research Experiences in math — January 2026

A tale of two densities: covering numbers and abundant numbers
- INTEGERS Conference, University of Georgia (Plenary) — May 2025
- Combinatorial and Additive Number Theory, CUNY — May 2025
- AMS Western Sectional Meeting, University of Denver — August 2025
- JMM - Session on Number Theory with Undergraduate Contributions — January 2026

From circle time to prime time
- JMM - TPSE Session Models for Inclusive Student Experiences, Seattle WA — January 2025

Knots and the Diaconis Graham Inequality for Permutations
- AMS Sectional Meeting, Howard University, Washington DC — April 2024

Short Interval Results For Powerfree Polynomials Over Finite Fields
- JMM, San Francisco CA — January 2024
- Canadian Number Theory Association Meeting, U. of Toronto — June 2024

The distribution of intermediate prime factors
- INTEGERS, Athens GA — May 2023
- Combinatorial and Additive Number Theory, CUNY — May 2023
- Journées Arithmétiques, University of Luxembourg — July 2025

Permutations and the divisor graph of $[1, n]$
- Number Theory Seminar, University of Georgia — October 2022
- JMM Special Session- Number Theory at non-PhD institutions, Boston MA — January 2023
- Colloquium, Loyola University Baltimore — March 2023
- Number Theory Seminar, Dartmouth College — May 2023

Primitive sets in function fields
- Combinatorial and Additive Number Theory (CUNY/Online) — June 2020

Counting pattern-avoiding integer partitions
- PAlmetto Number Theory Series, Clemson University — December 2019
- Mid Atlantic Seminar On Numbers, Gettysburg College — March 2020

Two combinatorial problems regarding primitive sets of integers
- Combinatorics Seminar, George Washington University — April 2019

Counting primitive sets and other statistics of the divisor graph of $\{1, 2, \ldots n\}$
- Combinatorial and Additive Number Theory, CUNY — May 2018
- INTEGERS, Augusta GA — October 2018

Unknotted Cycles
- Permutation Patterns, Dartmouth College — July 2018

Primitive and geometric-progression-free sets without large gaps
- Colloquium, University of Denver — April 2017
- Combinatorial and Additive Number Theory, CUNY — May 2017
- Canadian Number Theory Association Meeting XV, Laval University — July 2018
- MASON, James Madison University — February 2019
- Colloquium, Yale University — July 2019

Random multiplicative walks on the integers modulo n

Canadian Number Theory Association Meeting XIV, U. of Calgary	June 2016
INTEGERS, U. of West Georgia	October 2016
JMM Special Session on Analytic Number Theory, Atlanta GA	January 2017

Numbers divisible by a large shifted prime

SouthEast Regional Meeting On Numbers, James Madison University	April 2016
Combinatorial and Additive Number Theory, CUNY	May 2016

The convex hull of the prime number graph

Combinatorial and Additive Number Theory, CUNY	May 2015
Elementary, analytic, and algorithmic number theory, U. of Georgia	June 2015
Illinois Number Theory Conference, UIUC	August 2015

Popular values of the largest prime divisor function

Combinatorial and Additive Number Theory, CUNY	May 2014
Canadian Number Theory Association Meeting XIII, Carleton College	June 2014
Quebec/Maine Number Theory Conference, Université Laval	September 2014
Department Colloquium, University of Maine	October 2014
Joint Mathematics Meetings, Austin TX	January 2015
Southeastern AMS Sectional Meeting, Huntsville, AL	April 2015
Penn State Number Theory Seminar, Penn State University	April 2016

Unconventional Results in Multiplicative Combinatorial Number Theory

Invited Graduate Speaker, SERMON, Wofford College	April 2014

Using congruences to cover the integers

Graduate Student Seminar, Dartmouth College	February 2014

Things you can prove with a degree from DU, two results in number theory

Department Colloquium, University of Denver	February 2014

On sets of integers which contain no three terms in geometric progression

Maine/Quebec Number Theory Conference, University of Maine	October 2013
INTEGERS, University of West Georgia	October 2013
West Coast Number Theory, Pacific Grove, CA	December 2013
Joint Math Meetings, Baltimore, MD	January 2014
Exciting New Faces in Analytic Number Theory, Hausdorff Center for Mathematics, Bonn Germany	June 2014
Department Colloquium, Williams College	July 2014

When does each prime dividing $\varphi(n)$ also divide $n-1$?

Quebec/Maine Number Theory Conference, Université Laval	October 2012
Canadian Mathematics Society Winter Meeting, Montreal	December 2012

Probabilistic Galois Theory

Graduate Student Seminar, Dartmouth College	October 2011

Efficient realization of nonzero spectra by polynomial matrices

Graduate Student Seminar, Dartmouth College	October 2010
Departmental Colloquium, University of Denver	May 2010